

CSS — Fundamentals

Turn HTML structure into a designed interface.

CSS controls how HTML looks: colors, spacing, size, layout, and visual behavior.

background-color

```
background-color: #0d1117;
```

WHAT HAPPENS?

The page background changes to the chosen color.

TIP: Use hex, rgb(), or a named color.

color

```
color: white;
```

WHAT HAPPENS?

The text color of the selected element becomes white.

TIP: It inherits to many child elements.

font-size

```
font-size: 40px;
```

WHAT HAPPENS?

The selected text is displayed at a larger 40px size.

TIP: Use rem for scalable UI text.

style.css

CSS

```
1 body {  
2   background-color: #0d1117;  
3   color: white;  
4 }  
5  
6 h1 {  
7   font-size: 40px;  
8   margin: 0;  
9 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.

Styled website text

The browser applies the CSS here.

KEY TERMS

Selector

Chooses which HTML elements CSS will style.

Property

The feature you are changing, such as color.

Value

The setting given to a property.

CSS — Selectors & Colors

Target exactly the HTML elements you want to change.

Selectors tell the browser where a rule should apply.

Element selector

```
p { color: white; }
```

WHAT HAPPENS?

Every <p> element receives the rule.

TIP: Good for broad default styling.

Class selector

```
.card { ... }
```

WHAT HAPPENS?

Every element with class="card" receives the rule.

TIP: Classes are reusable.

ID selector

```
#title { ... }
```

WHAT HAPPENS?

The element with id="title" receives the rule.

TIP: Prefer classes for reusable styles.

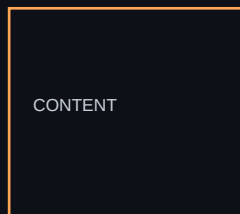
style.css

CSS

```
1 p { color: #e6edf3; }
2 .card { background: #161b22; }
3 #title { color: #58a6ff; }
4 button:hover {
5     background: #7c3aed;
6 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.



KEY TERMS

Selector

Pattern that chooses the element to style.

Class

Reusable label written with a leading dot in CSS.

Pseudo-class

A special state such as :hover.

CSS — Text & Fonts

Make content easier to read and give the interface a visual hierarchy.

Typography uses several CSS properties together: family, size, weight, alignment, and line spacing.

font-family

```
font-family: Arial, sans-serif;
```

WHAT HAPPENS?

The browser uses Arial first, then a fallback font.

TIP: Always give a fallback family.

font-weight

```
font-weight: 700;
```

WHAT HAPPENS?

The text becomes bold/heavier.

TIP: 400 is normal; 700 is bold.

line-height

```
line-height: 1.6;
```

WHAT HAPPENS?

More vertical space appears between lines of text.

TIP: Useful for comfortable reading.

style.css

CSS

```
1 body {  
2   font-family: Arial, sans-serif;  
3   line-height: 1.6;  
4 }  
5 h1 {  
6   font-size: 42px;  
7   font-weight: 700;  
8   text-align: center;  
9 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.

Styled website text

The browser applies the CSS here.

KEY TERMS

Typography

The visual styling and arrangement of text.

Fallback

A backup font used if the first choice is unavailable.

Hierarchy

Visual difference that shows what is most important.

CSS — The Box Model

Every element behaves like a box with content, padding, border, and margin.

The box model is one of the most important CSS concepts because spacing and size depend on it.

padding

`padding: 20px;`

WHAT HAPPENS?

Space is added inside the element, around its content.

TIP: Think: inside space.

border

`border: 1px solid #30363d;`

WHAT HAPPENS?

A visible boundary is drawn around the box.

TIP: You can control width, style, and color.

margin

`margin: 20px;`

WHAT HAPPENS?

Space is added outside the element, separating it from others.

TIP: Think: outside space.

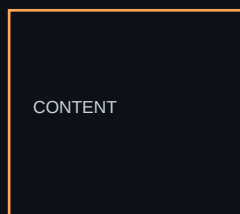
style.css

CSS

```
1 .card {  
2   width: 300px;  
3   padding: 20px;  
4   border: 1px solid #30363d;  
5   margin: 20px;  
6   box-sizing: border-box;  
7 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.



KEY TERMS

Attribute

Extra information written inside an opening tag.

Accessibility

Making content usable by more people.

Media

Visual or audio content included in a page.

CSS — Display & Flexbox

Control how elements sit next to each other and align inside containers.

Flexbox is the everyday layout tool for rows, columns, alignment, and spacing.

display: flex

```
display: flex;
```

WHAT HAPPENS?

The children become flex items and can be arranged as a flexible row/column.

TIP: Start here for most simple layouts.

justify-content

```
justify-content: space-between;
```

WHAT HAPPENS?

Items are distributed along the main axis with space between them.

TIP: Main-axis alignment.

align-items

```
align-items: center;
```

WHAT HAPPENS?

Items are centered across the cross axis.

TIP: Cross-axis alignment.

style.css

CSS

```
1 .container {  
2   display: flex;  
3   justify-content: space-between;  
4   align-items: center;  
5   gap: 16px;  
6   flex-wrap: wrap;  
7 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.

ITEM 1

ITEM 2

KEY TERMS

Box model

How an element's content and surrounding space are calculated.

Content

The actual text, image, or child content inside the element.

Border

The visible edge between an element and its outside area.

CSS — Positioning

Move elements with control and understand what they are positioned against.

Positioning changes how an element participates in the normal page flow.

position: relative

```
position: relative;
```

WHAT HAPPENS?

The element stays in normal flow and becomes a reference point for positioned children.

TIP: Often paired with absolute.

position: absolute

```
position: absolute; top: 12px;
```

WHAT HAPPENS?

The element is removed from normal flow and positioned against its nearest positioned ancestor.

TIP: Use carefully for overlays.

z-index

```
z-index: 10;
```

WHAT HAPPENS?

The positioned element can appear above elements with lower stacking order.

TIP: Works within stacking contexts.

style.css

CSS

```
1 .badge {  
2   position: absolute;  
3   top: 12px;  
4   right: 12px;  
5 }  
6 .page {  
7   position: relative;  
8 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.



KEY TERMS

Flex container

The parent element with display: flex.

Main axis

The direction in which flex items flow.

Cross axis

The axis perpendicular to the main axis.

CSS — Borders, Shadows & Hover

Add depth and interaction without making the design noisy.

Small visual effects can make cards and buttons feel polished.

border-radius

```
border-radius: 14px;
```

WHAT HAPPENS?

The corners of the element become rounded.

TIP: Larger values create softer corners.

box-shadow

```
box-shadow: 0 8px 24px ...;
```

WHAT HAPPENS?

A shadow appears around the element, creating depth.

TIP: Keep shadows subtle.

transition

```
transition: transform .2s ease;
```

WHAT HAPPENS?

CSS changes animate smoothly instead of switching instantly.

TIP: Pair with :hover for interaction.

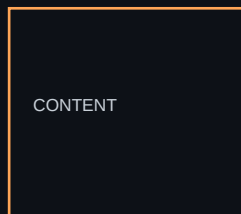
style.css

CSS

```
1 .card {  
2   border-radius: 14px;  
3   box-shadow: 0 8px 24px rgba(0,0,0,.25);  
4   transition: transform .2s ease;  
5 }  
6 .card:hover {  
7   transform: translateY(-4px);  
8 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.



KEY TERMS

Transition

Smooth change between CSS states.

Hover

The state when a pointer is over an element.

Transform

A visual move, scale, rotate, or skew of an element.

CSS — Responsive Design

Make the same website work across desktop, tablet, and mobile screens.

Responsive CSS adapts layout and sizing to the available viewport.

max-width

```
max-width: 700px;
```

WHAT HAPPENS?

Limits how wide an element can become.

TIP: Useful for readable content widths.

@media

```
@media (max-width: 700px) { ... }
```

WHAT HAPPENS?

Applies styles only when the screen matches the condition.

TIP: This is the core of responsive rules.

1fr

```
grid-template-columns: 1fr;
```

WHAT HAPPENS?

Gives one equal fraction of available grid space to the column.

TIP: Great for flexible layouts.

style.css

CSS

```
1 .grid {  
2   display: grid;  
3   grid-template-columns: repeat(3, 1fr);  
4   gap: 16px;  
5 }  
6 @media (max-width: 700px) {  
7   .grid { grid-template-columns: 1fr; }  
8 }
```

BROWSER RESULT

This is the visible effect you should expect after the browser applies the CSS rule.

ITEM 1

ITEM 2

KEY TERMS

Viewport

The visible area of the browser window.

Media query

A CSS condition that applies styles for specific situations.

Responsive

A layout that adapts to different screen sizes.